

Notes on Markov Decision Processes

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DISCOUNT FACTOR

The discount factor γ is a parameter in reinforcement learning that determines the importance of future rewards compared to immediate rewards. It is a value between 0 and 1, where: - A value of 0 means that the agent only cares about immediate rewards and ignores future rewards. We want to maximize the total discounted reward, which is given by the formula:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Where G_t is the total discounted reward at time step t , and R_{t+k+1} is the reward received at time step $t+k+1$.

POLICY $\pi(s)$

A policy is a strategy that the agent follows to determine which action to take in each state. It is a mapping from states to actions, and is denoted by $\pi(a|s)$, which represents the probability of taking action a in state s .

stochastic policy: $\pi(a|s)$ is a probability distribution over actions given a state. It allows for randomness in action selection, which can be useful for exploration. deterministic policy: $\pi(s)$ is a function that maps each state to a specific action. It does not allow for randomness, and the agent will always take the same action in a given state.

We want to maximize the sum of discounted rewards, which can be expressed as:

$$J(\pi) = \sum_s d^\pi(s) \sum_a \pi(a|s) Q^\pi(s, a)$$

Where $J(\pi)$ is the expected return of the policy π , $d^\pi(s)$ is the stationary distribution of states under policy π , and $Q^\pi(s, a)$ is the action-value function, which represents the expected return of taking action a in state s and following policy π thereafter.

We must find the optimal policy π^* that maximizes $J(\pi)$, which can be expressed as:

$$\pi^* = \arg \max_{\pi} J(\pi)$$

STATE VALUE

The state value function, denoted as $V^\pi(s)$, represents the expected return (total discounted reward) when starting from state s and following policy π thereafter. It can be defined as:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s \right]$$

Where $V^\pi(s)$ is the state value function for state s under policy π , and the expectation is taken over the distribution of rewards and states that result from following policy π starting from state s .

STATE-ACTION VALUE

The state-action value function, denoted as $Q^\pi(s, a)$, represents the expected return (total discounted reward) when starting from state s , taking action a , and following policy π thereafter. It can be defined as:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a \right]$$

Where $Q^\pi(s, a)$ is the state-action value function for state s and action a under policy π , and the expectation is taken over the distribution of rewards and states that result from following policy π starting from state s and taking action a .

BELLMAN EQUATIONS

The Bellman equations are fundamental recursive relationships that relate the value of a state or state-action pair to the values of its successor states or state-action pairs. They are used to compute the value functions and to derive optimal policies. The Bellman equation for the state value function is given by:

$$V^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V^\pi(s')]$$

Where $P(s'|s, a)$ is the transition probability of moving from state s to state s' after taking action a , and $R(s, a, s')$ is the reward received for taking action a in state s and transitioning to state s' . The Bellman equation for the state-action value function is given by:

$$Q^\pi(s, a) = \sum_{s'} P(s'|s, a) \left[R(s, a, s') + \gamma \sum_{a'} \pi(a'|s') Q^\pi(s', a') \right]$$

Where the terms are defined as in the previous equation.

DECISION PROCESS

A Markov Decision Process (MDP) is a mathematical framework for modeling decision-making problems where outcomes are partly random and partly under the control of a decision-maker. An MDP is defined by the following components: - A set of states S : This represents all possible states that the system can be in

The optimal value of a state is the expected return when starting from that state and following the optimal policy thereafter. It can be defined as:

$$V^*(s) = \max_{\pi} V^\pi(s)$$

Where $V^*(s)$ is the optimal state value function for state s , and the maximum is taken over all possible policies π .

The optimal action-value function is the expected return when starting from a state, taking an action, and following the optimal policy thereafter. It can be defined as:

$$Q^*(s, a) = \max_{\pi} Q^{\pi}(s, a)$$

Where $Q^*(s, a)$ is the optimal state-action value function for state s and action a , and the maximum is taken over all possible policies π .